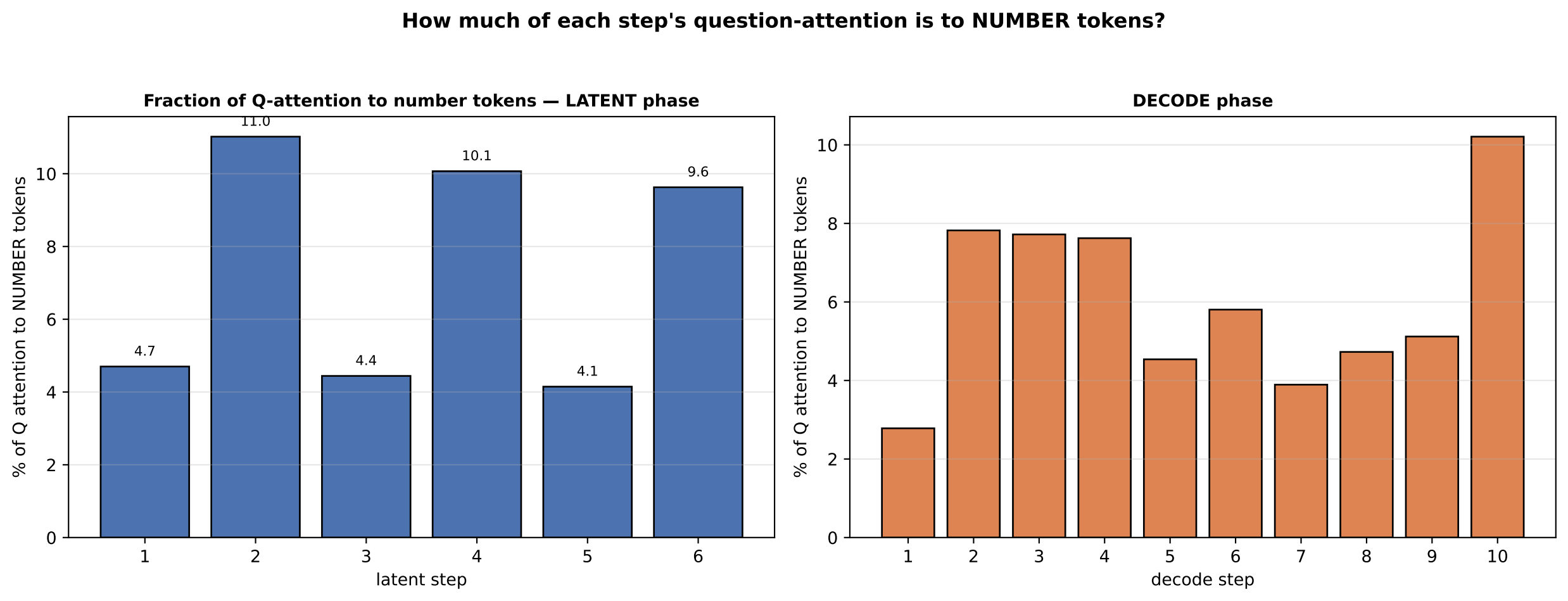
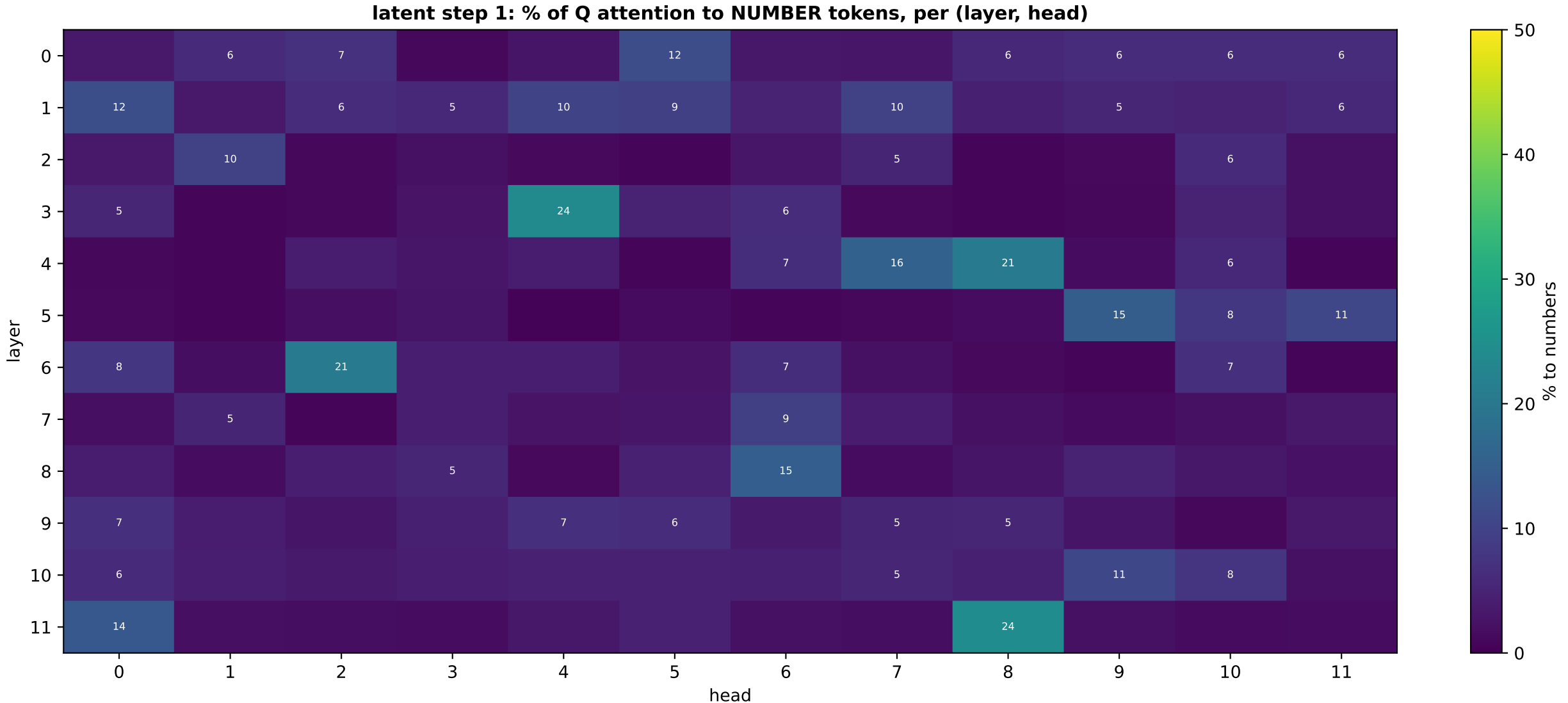
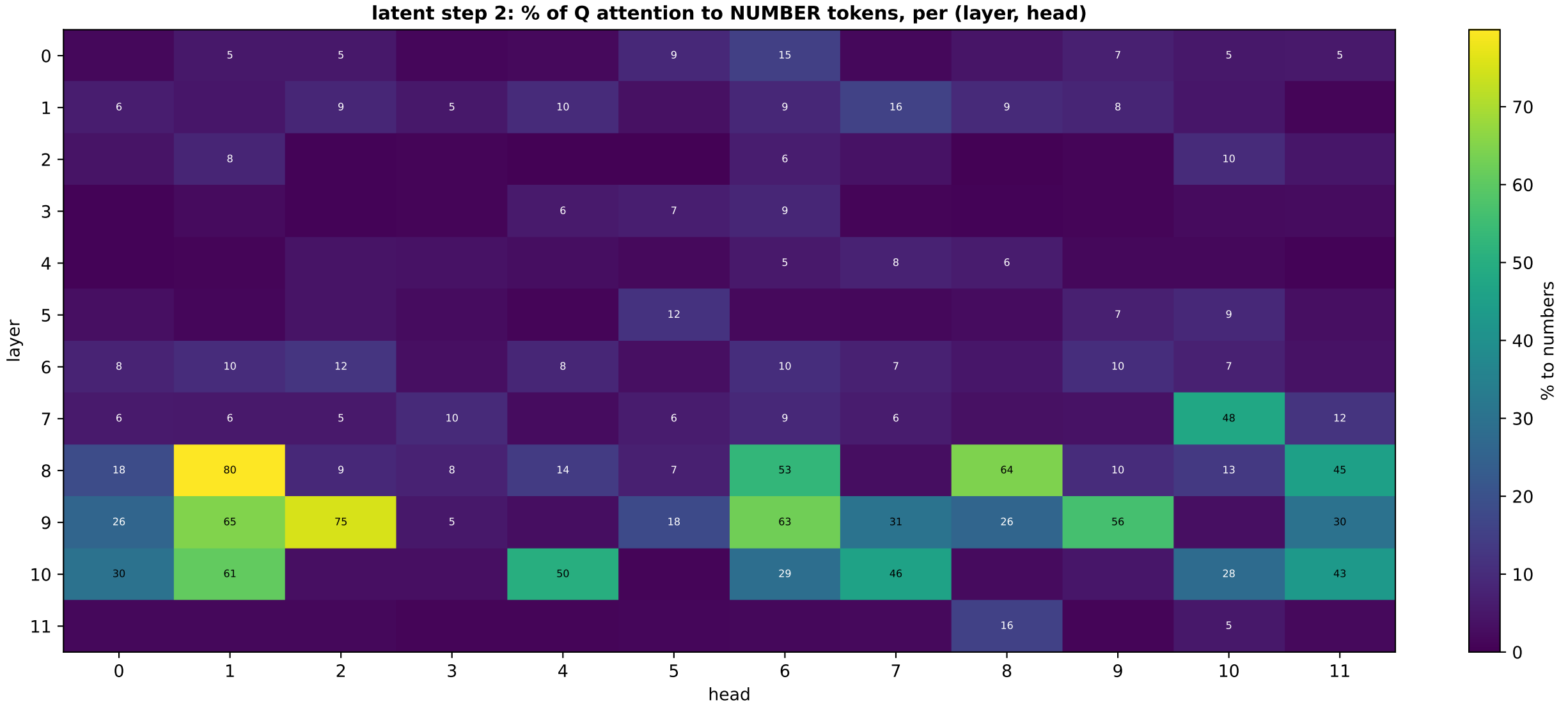
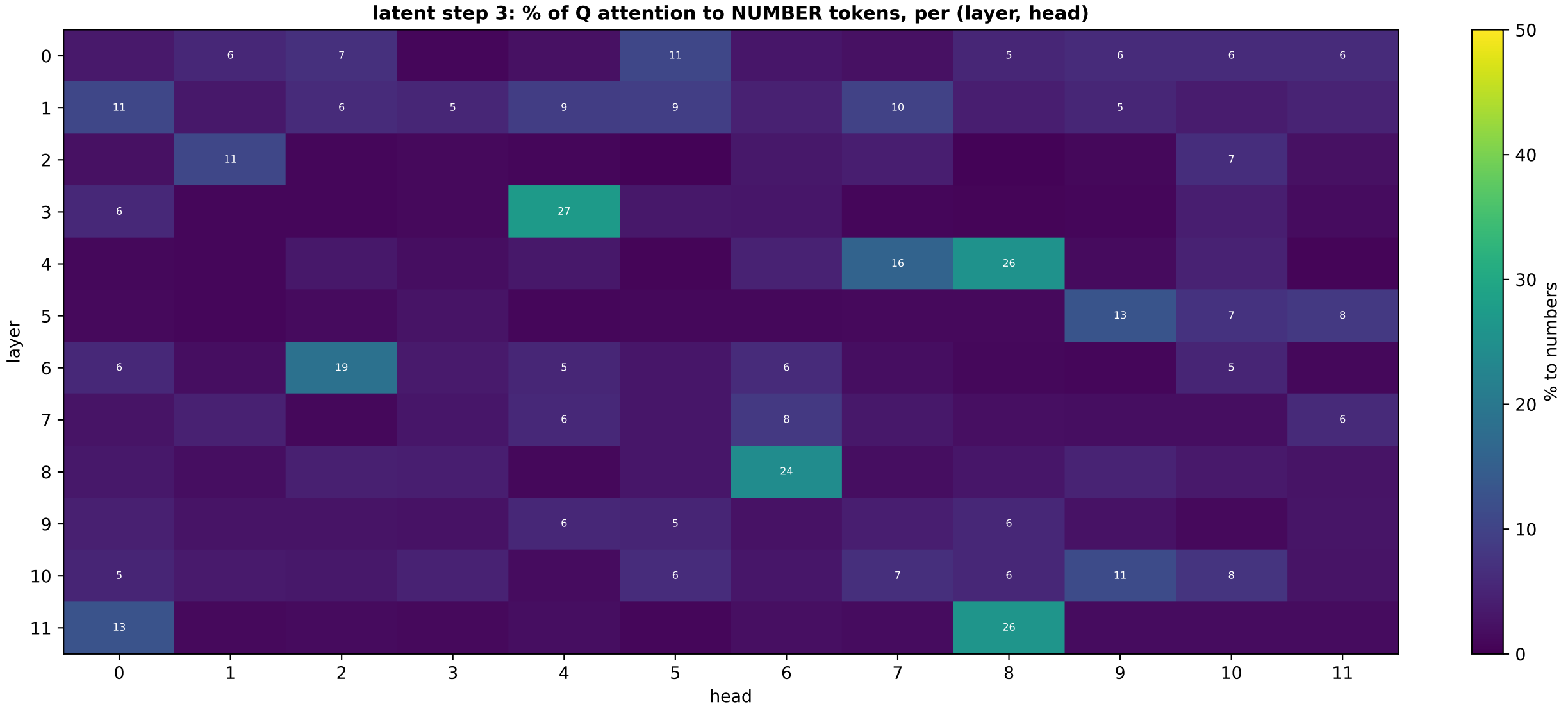
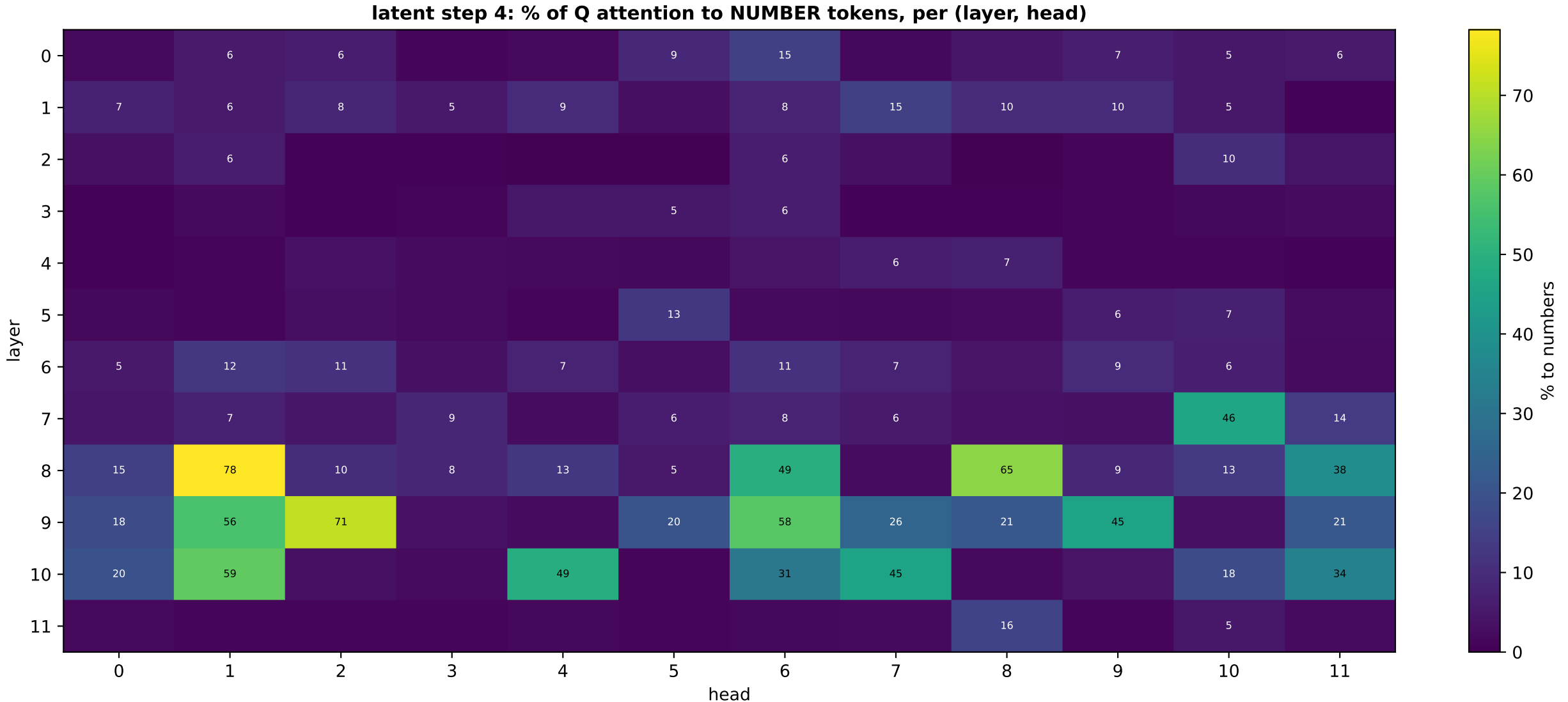
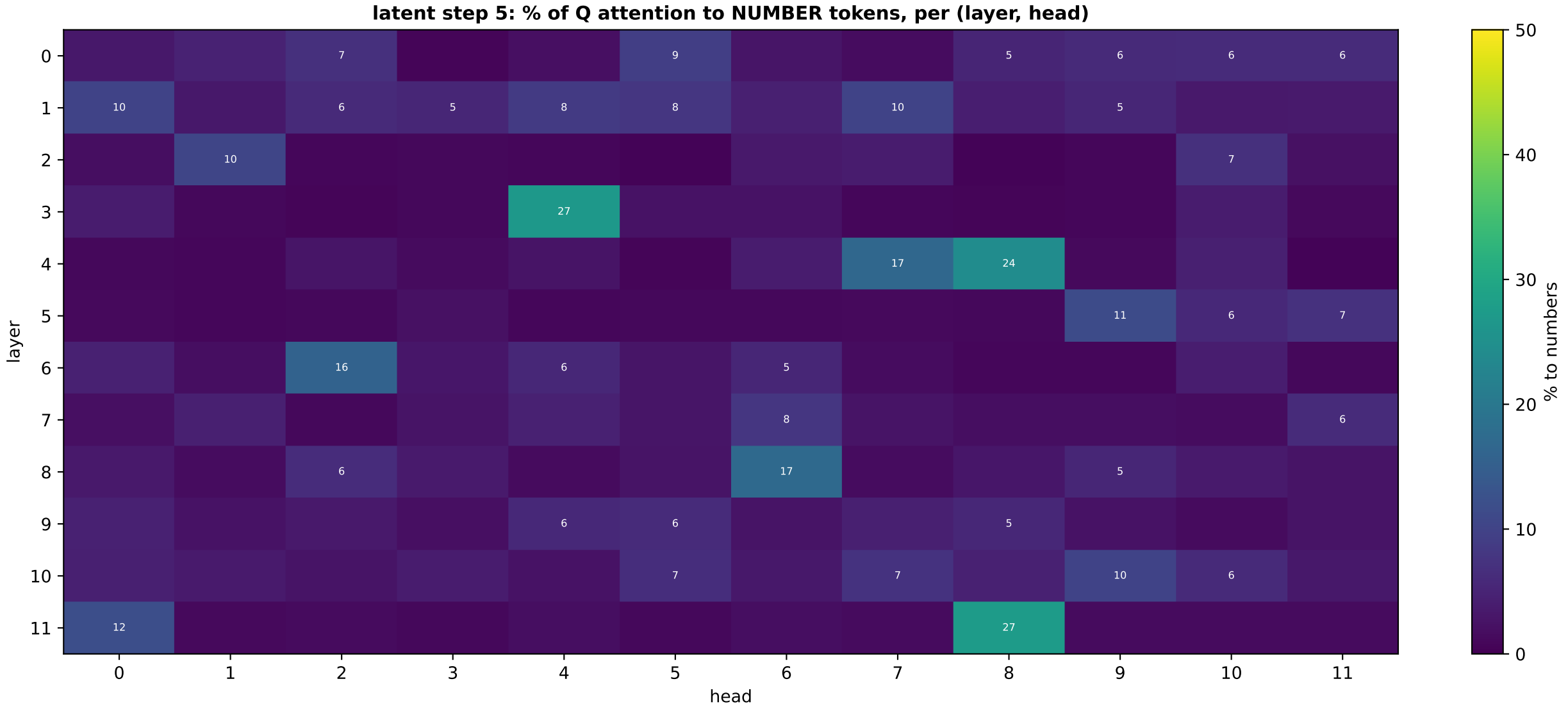

How much of each step's question-attention is to NUMBER tokens?
Fraction of Q-attention to number tokens — LATENT phase
% of Q attention to NUMBER tokens
latent step
4.7
11.0
4.4
10.1
4.1
9.6
DECODE phase
% of Q attention to NUMBER tokens
decode step

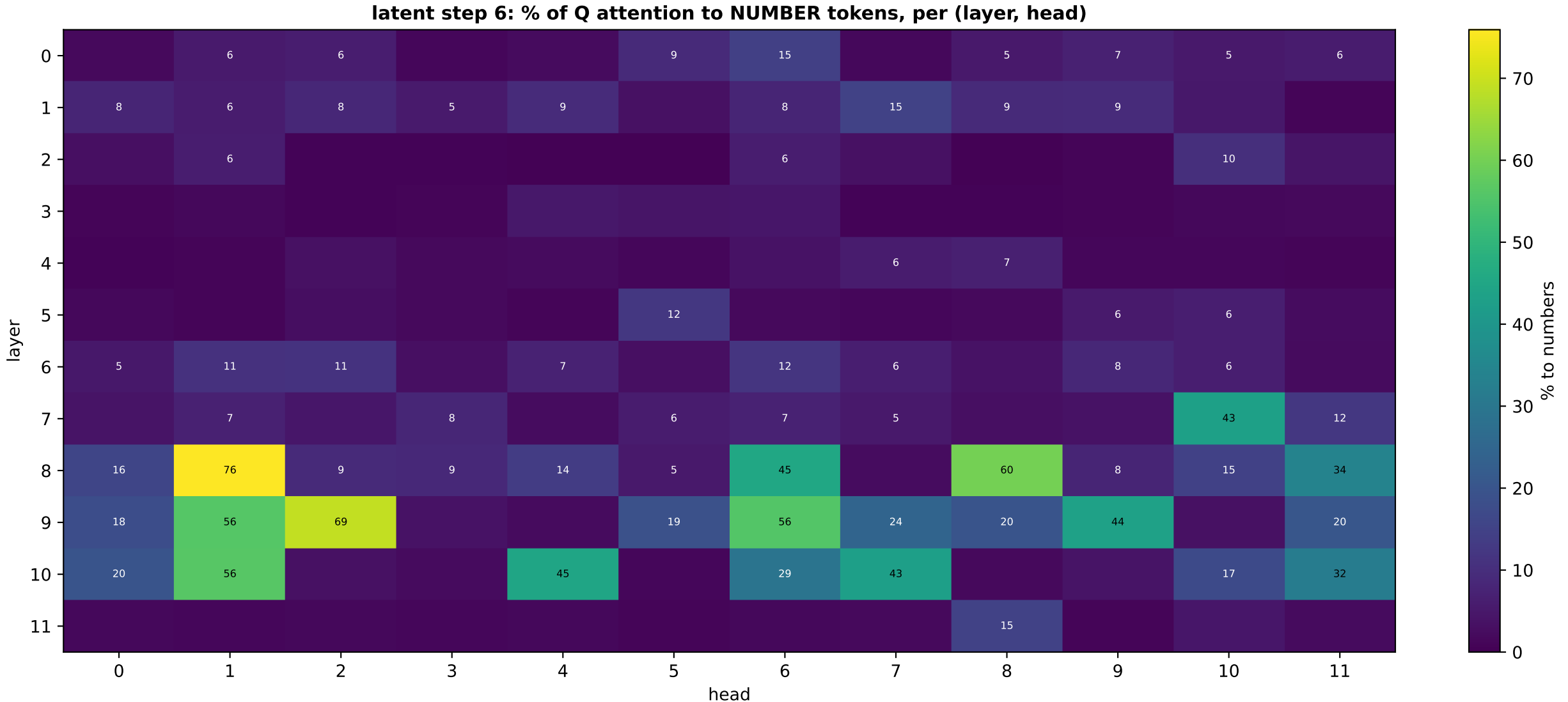

latent step 1: % of Q attention to NUMBER tokens, per (layer, head)
layer
head
% to numbers
0 1 2 3 4 5 6 7 8 9 10 11
0 1 2 3 4 5 6 7 8 9 10 11
0 10 20 30 40 50
0: 6 7 12 6 6 6 6
1: 12 6 5 10 9 10 5 6
2: 10 5 6
3: 5 24 6
4: 7 16 21 6
5: 15 8 11
6: 8 21 7 7
7: 5 9
8: 5 15
9: 7 7 6 5 5
10: 6 5 11 8
11: 14 24

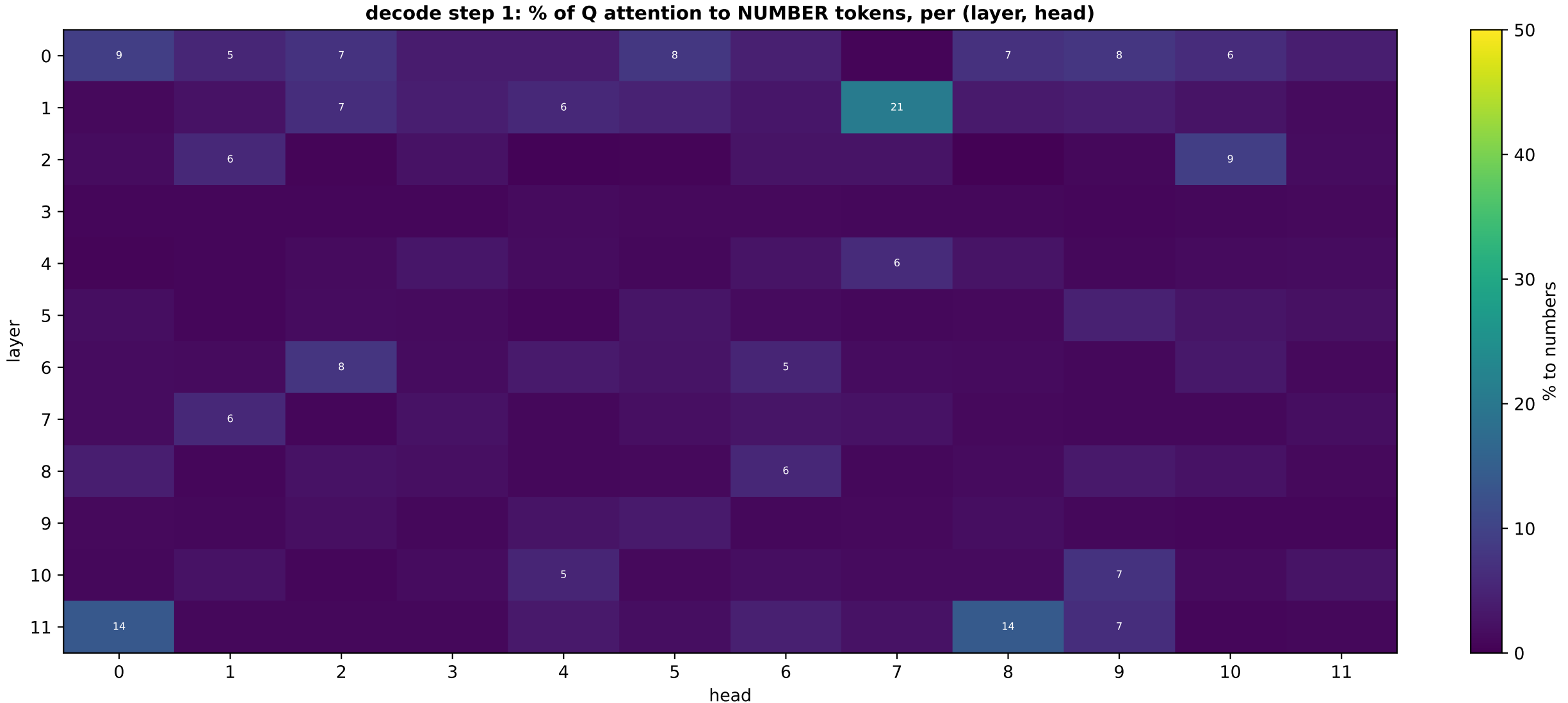

latent step 2: % of Q attention to NUMBER tokens, per (layer, head)
layer
head
% to numbers
0 1 2 3 4 5 6 7 8 9 10 11
0: 5 5 9 15 7 5 5
1: 6 9 5 10 9 16 9 8
2: 8 6 10
3: 6 7 9
4: 5 8 6
5: 12 7 9
6: 8 10 12 8 10 7 10 7
7: 6 6 5 10 6 9 6 48 12
8: 18 80 9 8 14 7 53 64 10 13 45
9: 26 65 75 5 18 63 31 26 56 30
10: 30 61 50 29 46 28 43
11: 16 5
0 10 20 30 40 50 60 70

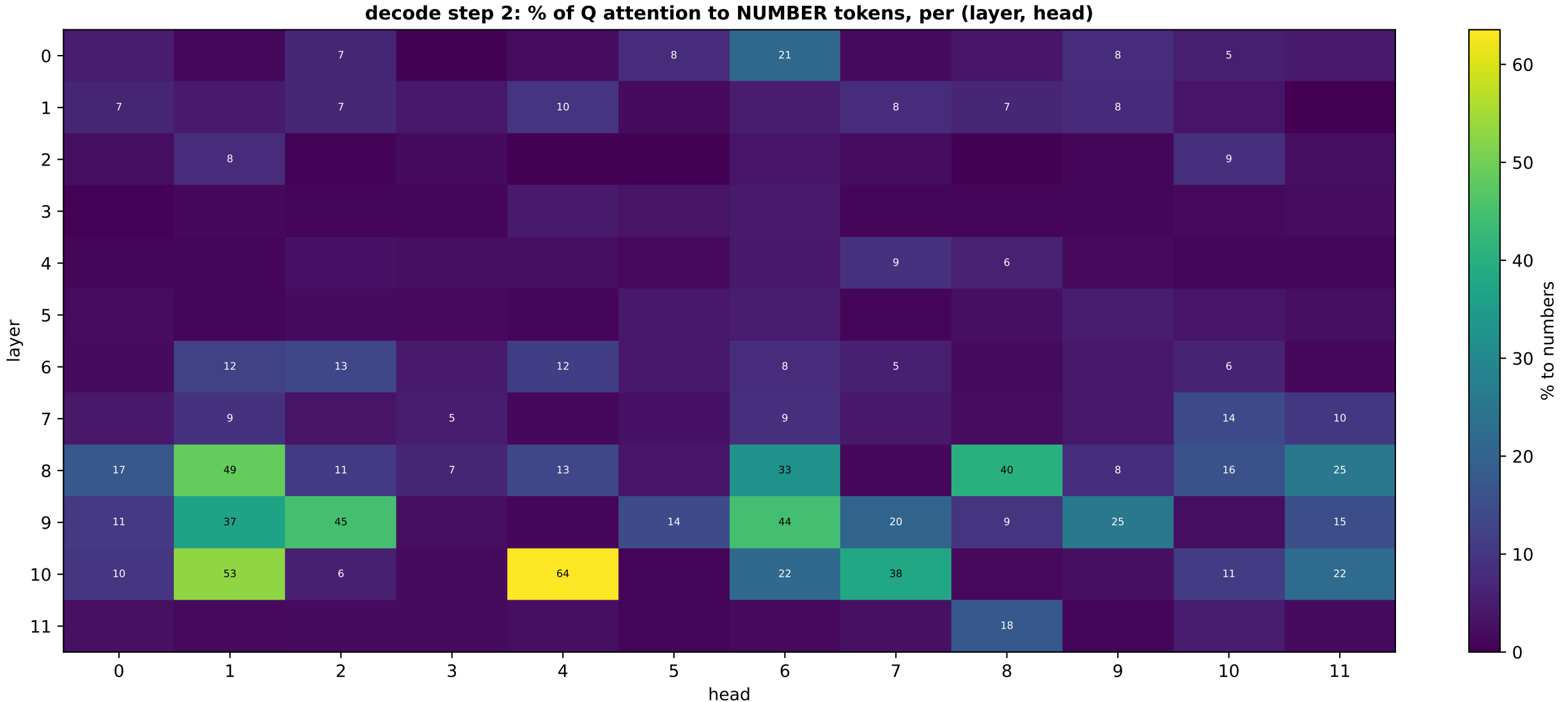

latent step 3: % of Q attention to NUMBER tokens, per (layer, head)
layer
head
% to numbers
0 1 2 3 4 5 6 7 8 9 10 11
0 1 2 3 4 5 6 7 8 9 10 11
0 10 20 30 40 50
0: 6 7 11 5 6 6 6
1: 11 6 5 9 9 10 5
2: 11 7
3: 6 27
4: 16 26
5: 13 7 8
6: 6 19 5 6 5
7: 6 8 6
8: 24
9: 6 5 6
10: 5 6 7 6 11 8
11: 13 26

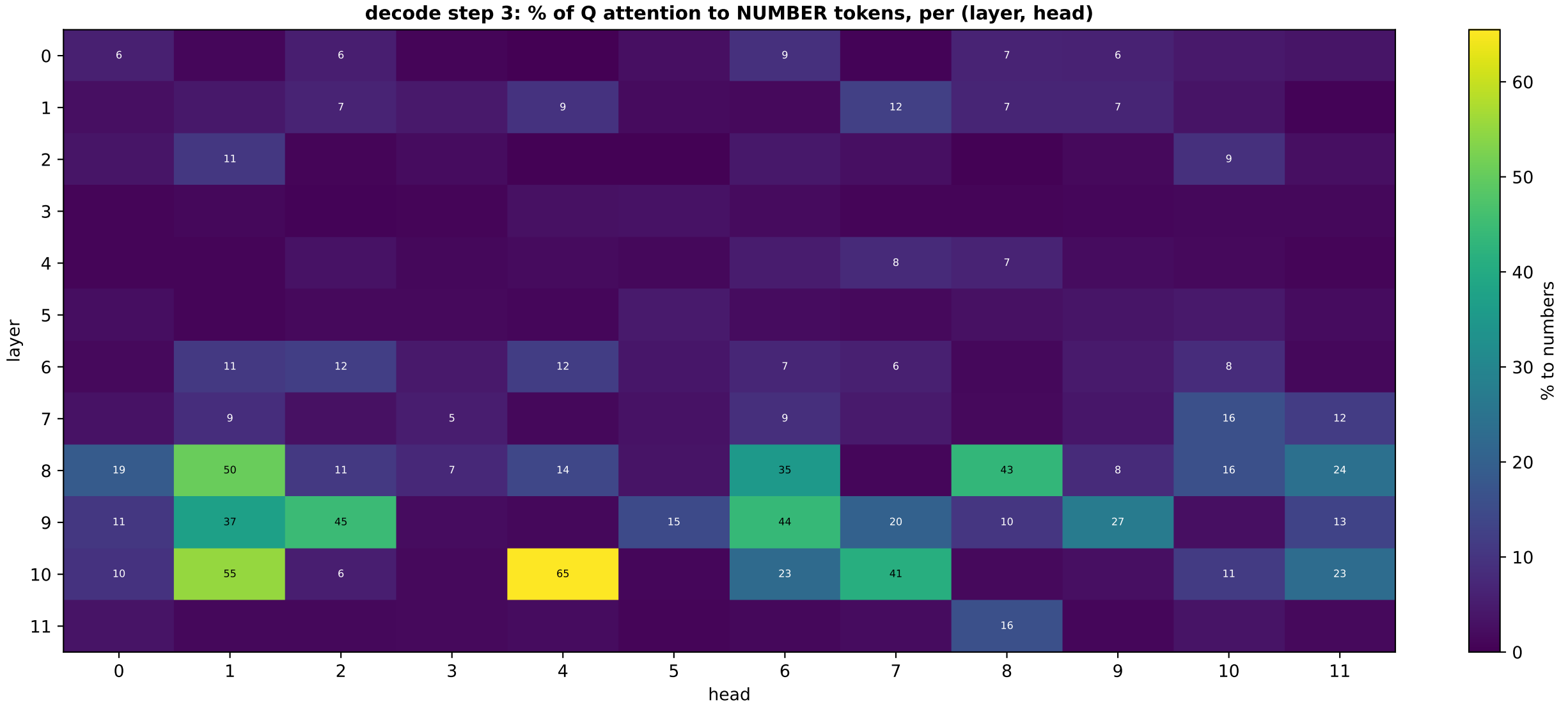

latent step 4: % of Q attention to NUMBER tokens, per (layer, head)
layer
head
% to numbers
0 1 2 3 4 5 6 7 8 9 10 11
0 1 2 3 4 5 6 7 8 9 10 11
0 10 20 30 40 50 60 70
0: 6 6 9 15 7 5 6
1: 7 6 8 5 9 8 15 10 10 5
2: 6 6 10
3: 5 6
4: 6 7
5: 13 6 7
6: 5 12 11 7 11 7 9 6
7: 7 9 6 8 6 46 14
8: 15 78 10 8 13 5 49 65 9 13 38
9: 18 56 71 20 58 26 21 45 21
10: 20 59 49 31 45 18 34
11: 16 5

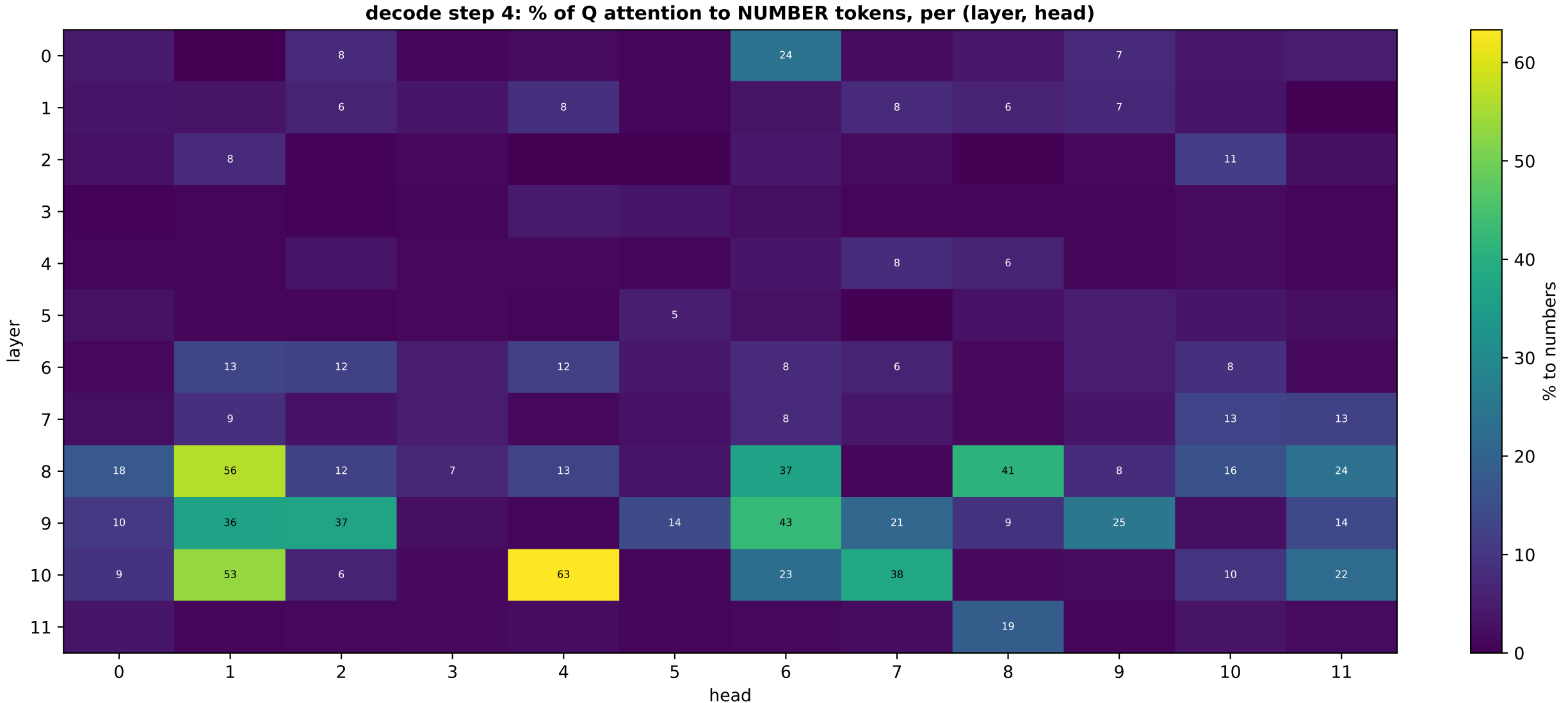

latent step 5: % of Q attention to NUMBER tokens, per (layer, head)
layer
head
% to numbers
0
1
2
3
4
5
6
7
8
9
10
11
12
16
17
24
27
50
40
30
20
10


latent step 6: % of Q attention to NUMBER tokens, per (layer, head)
layer
head
% to numbers
0 1 2 3 4 5 6 7 8 9 10 11
0 1 2 3 4 5 6 7 8 9 10 11
0 10 20 30 40 50 60 70
0: 6 6 9 15 5 7 5 6
1: 8 6 8 5 9 8 15 9 9
2: 6 6 10
4: 6 7
5: 12 6 6
6: 5 11 11 7 12 6 8 6
7: 7 8 6 7 5 43 12
8: 16 76 9 9 14 5 45 60 8 15 34
9: 18 56 69 19 56 24 20 44 20
10: 20 56 45 29 43 17 32
11: 15


decode step 1: % of Q attention to NUMBER tokens, per (layer, head)
layer
head
% to numbers
0
1
2
3
4
5
6
7
8
9
10
11
9
5
7
8
7
8
6
7
6
21
6
9
6
8
5
6
6
6
5
7
14
14
7
50
40
30
20
10
0


decode step 2: % of Q attention to NUMBER tokens, per (layer, head)
layer
head
% to numbers
0 1 2 3 4 5 6 7 8 9 10 11
0 1 2 3 4 5 6 7 8 9 10 11
0 10 20 30 40 50 60
7 8 21 8 5
7 7 10 8 7 8
8 9
9 6
12 13 12 8 5 6
9 5 9 14 10
17 49 11 7 13 33 40 8 16 25
11 37 45 14 44 20 9 25 15
10 53 6 64 22 38 11 22
18


decode step 3: % of Q attention to NUMBER tokens, per (layer, head)
layer
head
% to numbers
0 1 2 3 4 5 6 7 8 9 10 11
0 1 2 3 4 5 6 7 8 9 10 11
0 10 20 30 40 50 60
6 6 9 7 6
7 9 12 7 7
11 9
8 7
11 12 12 7 6 8
9 5 9 16 12
19 50 11 7 14 35 43 8 16 24
11 37 45 15 44 20 10 27 13
10 55 6 65 23 41 11 23
16


decode step 4: % of Q attention to NUMBER tokens, per (layer, head)
layer
head
% to numbers
0 1 2 3 4 5 6 7 8 9 10 11
0 1 2 3 4 5 6 7 8 9 10 11
0 10 20 30 40 50 60
8 24 7
6 8 8 6 7
8 11
8 6
5
13 12 12 8 6 8
9 8 13 13
18 56 12 7 13 37 41 8 16 24
10 36 37 14 43 21 9 25 14
9 53 6 63 23 38 10 22
19